An evening of exposure

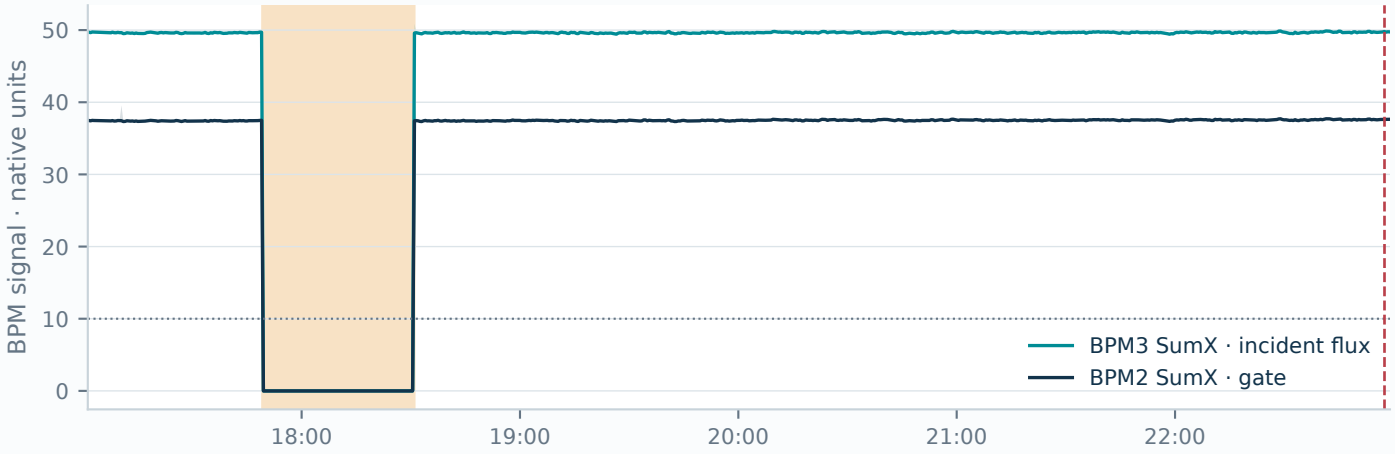
Kapton atmospheric-air holder | 6 keV | WAXS + on-axis imaging

4 h 59 min 22 s
BEST ESTIMATE · BEAM ON
Includes early alarm correction

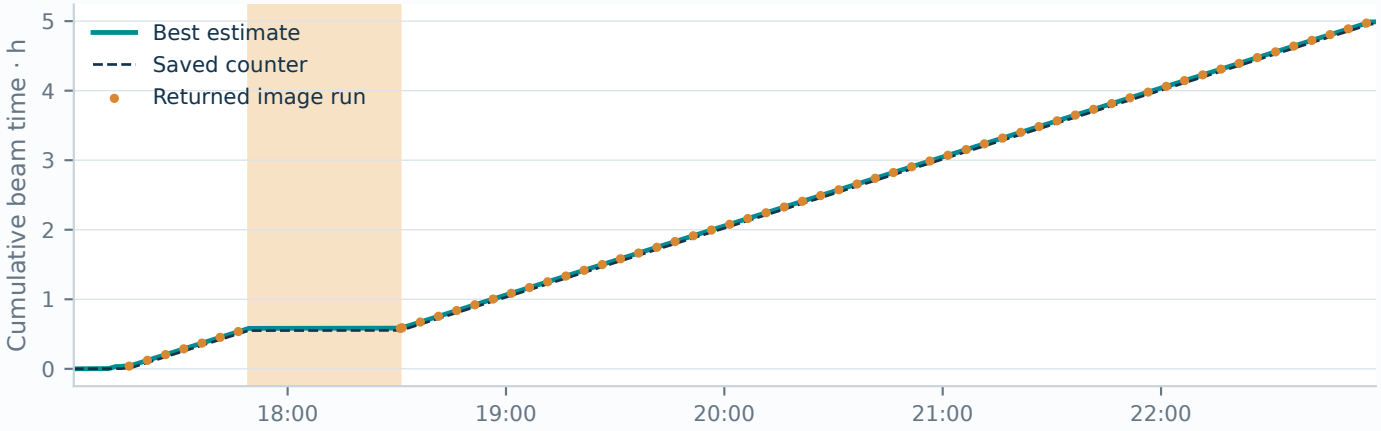
22:57:33
FINAL FAST-SHUTTER CLOSE
22 October · first closed sample

41 min 41 s
BEAM INTERRUPTION
17:49:15 → 18:30:56

01 Incident beam disappeared well before 18:30



02 Exposure accumulated; outage and shutter gaps excluded



THE RESULT

Saved counter: 4 h 57 min 26 s • Early correction: 116.0 s

Fast shutter open: 5 h 41 min 03 s • 62 image runs returned

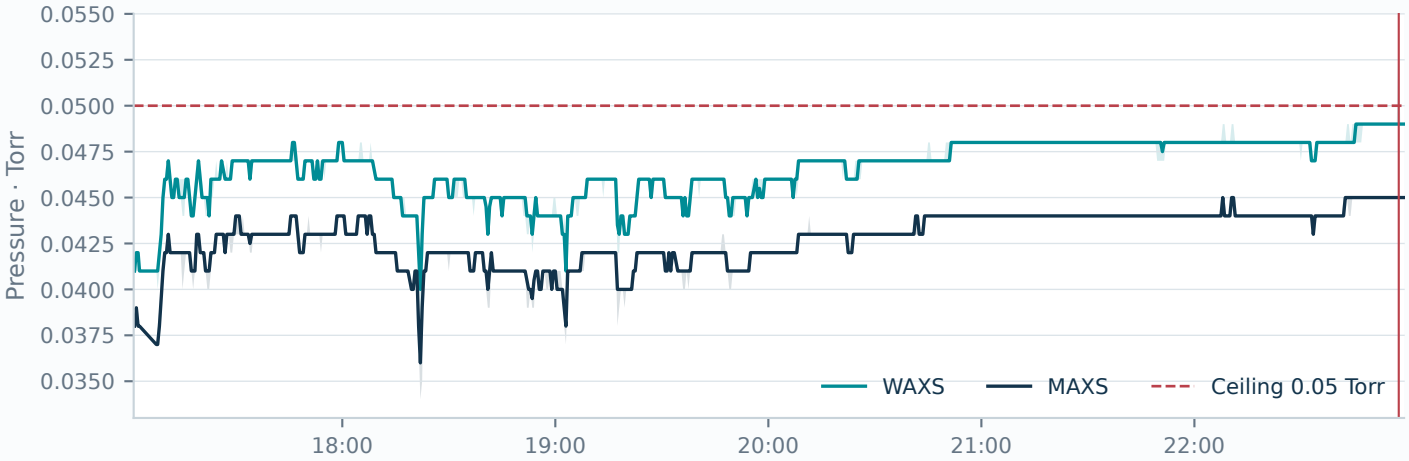
Beam loss started at 17:49:15; 18:30:56 is the return, not the loss onset. Both BPMs dropped while FE stayed open. This is compatible with the reported upstream valve trip; no gate-valve readback was logged to identify the valve.

Exposure is a shutter/BPM proxy, not calibrated absorbed dose. No time after the last sample or across unobserved restart gaps is added to the estimate.

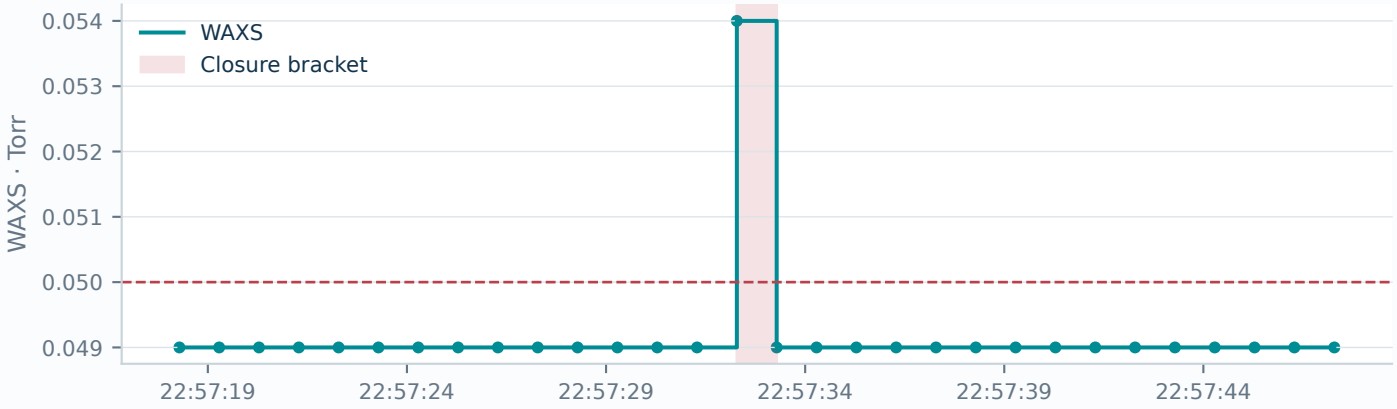
A short spike ended the run

Pressure evolution | WAXS and MAXS only | SAXS isolated, per operator

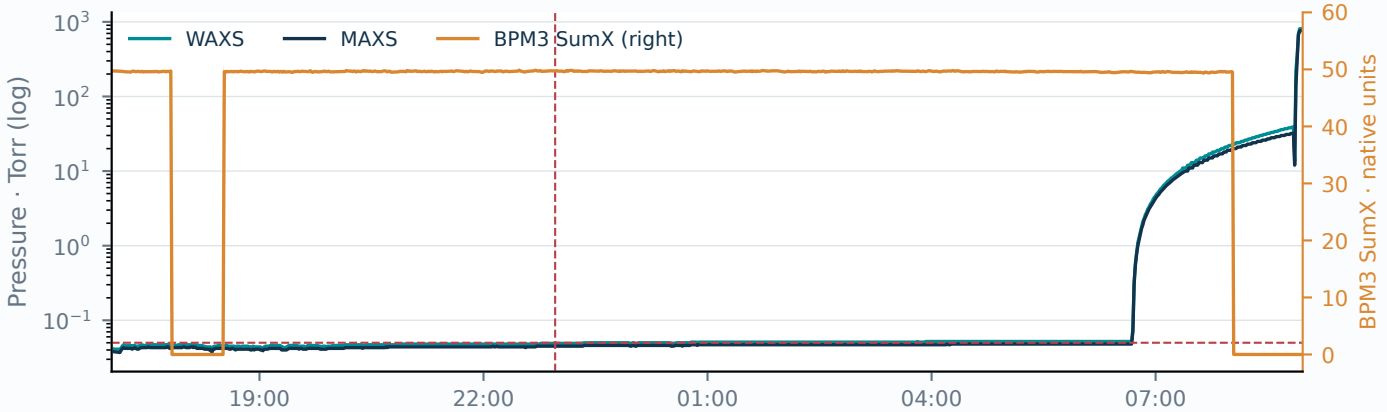
03 WAXS approached the ceiling through the evening



04 0.054 Torr for one logged sample; then shutter closed



05 Overnight pressure and incident flux



22:57:32.284: WAXS = 0.054 Torr; MAXS = 0.045 Torr.

22:57:33.285: fast shutter closed; WAXS had returned to 0.049 Torr.

Morning pressure rise occurred with the shutter closed; cause is not established.

Statistics & the audit trail

Measured totals, quality limits and what the image records establish

Best estimate of beam-on time	4 h 59 min 22 s
Saved online beam counter	4 h 57 min 26 s
Added early MINOR-alarm interval	116.03 s
Total fast-controller-open time	5 h 41 min 03 s
Open shutter, no detected beam	0 h 41 min 41 s
BPM3 beam-weighted mean ± SD	49.641 ± 0.264 native units
BPM3 relative variability (weighted CV)	0.53%
Integrated BPM3 × beam time	891,662 signal·s (not photons)
WAXS / MAXS median while acquiring beam	0.047 / 0.043 Torr
Image runs returned / attempts	62 / 63
Median image spacing	299.94 s
Logged samples / typical interval	57,135 / ~1 s
Unobserved restart time	282.51 s (not credited)

INTERPRETATION

- Final closure is bracketed by 22:57:32.284 and 22:57:33.285 EDT. Counter interpolation estimates 22:57:32.600; use ~1 s timing precision.
- The first WAXS threshold crossing immediately preceded closure. The sequence matches software pressure-stop behavior; no RunEngine traceback was supplied.
- All 62 returned image runs carry the conservative 'interrupted' flag. The detector's normal end-of-exposure shutter closure can set this flag; it is not evidence that all images are unusable. One early attempt failed.
- First returned image Oct 22, 17:16:26; last Oct 22, 22:56:27.

METHOD & LIMITS

Use the final cumulative counters, never sum their per-sample values. The early 116 s correction uses fast-open increments with BPM3 in 45–55 (near 49), FE=0, photon='Open'/MINOR and BPM2>10. This follows the operator's confirmed interpretation of beam on the Kapton despite the photon alarm. No downtime is recovered. Transition latency and sub-sample changes limit accuracy; downstream valve position and absorbed dose were not measured. BPM3 statistics weight fresh readings by corrected beam-time increments. Plots show 30–60 s medians and min–max envelopes; spike panel uses raw data. SAXS pressure is omitted: the branch was isolated and zero was below range. Image counts come from returned run UIDs; before/after frames are on page 4.

NEXT SEGMENT: leak repair; SAXS-connected operation if vacuum permits.
Backup: WAXS isolated, ~0.10 Torr trip. Add close-only chamber/pump isolation.

Source: kapton_air_holder_01_bpm2_bpm3_waxs_oav.sqlite
SHA-256: 74b1a2976cbc8f9dc849fbb6edcc70ad2326f9f5…
Coverage: Oct 22, 17:01:14 → Oct 23, 08:58:10 EDT · specimen kapton_air_holder_01

Before & after exposure

Matched scale | Earliest and latest returned WAXS + OAV frames

BEFORE

Oct 22, 17:16:26 EDT

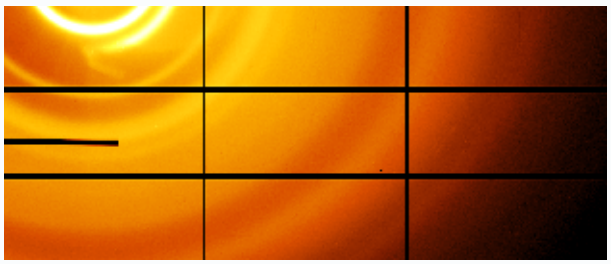
Beam on: 0 h 02 min 21 s

AFTER

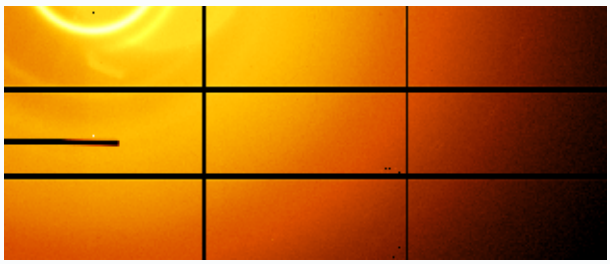
Oct 22, 22:56:27 EDT

Beam on: 4 h 58 min 19 s

WAXS · scan 1185420



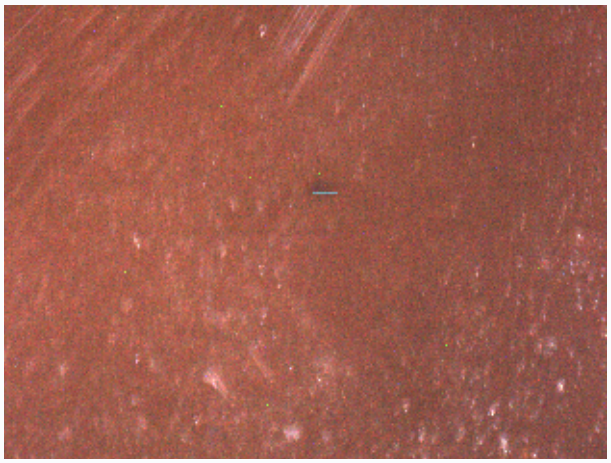
WAXS · scan 1185481



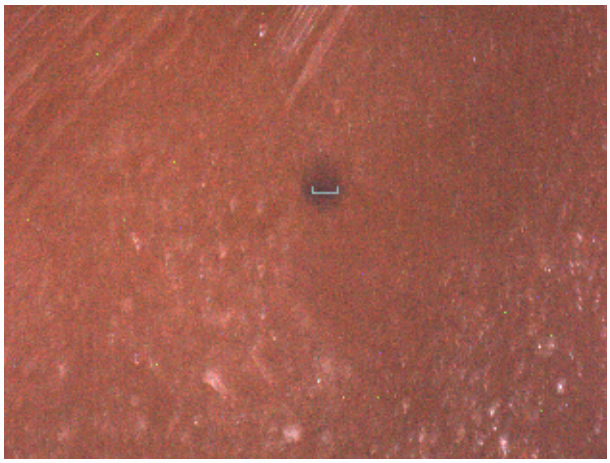


WAXS counts · fixed logarithmic yellow-hot scale

On-axis view



On-axis view



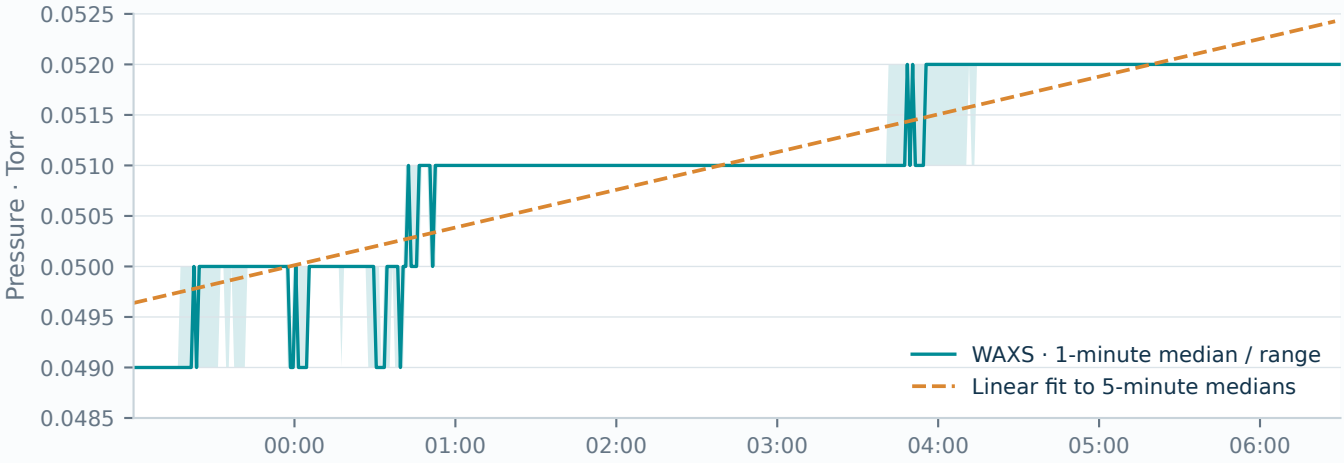
'Before' means first successfully returned frame, not an unirradiated reference. The final frame precedes shutter closure by about 66 seconds. Negative/zero WAXS pixels are masked. Movies use this same fixed scale and annotated totals.

The slow overnight rise

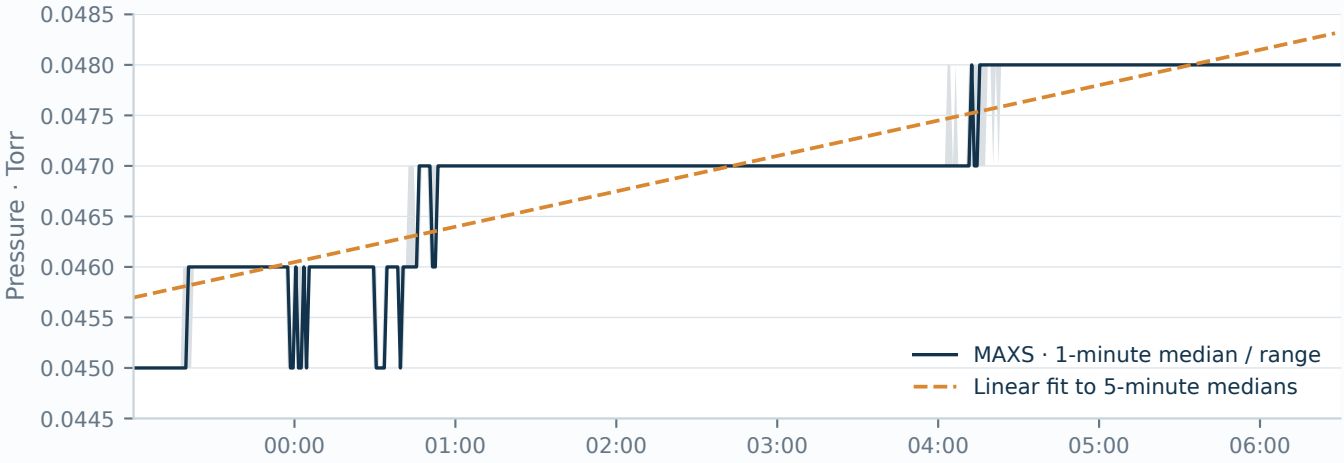
Fast shutter closed throughout | Linear pressure scale | Before the rapid morning rise

23:00 EDT, Oct 22 → 06:30 EDT, Oct 23 · 7.5 hours

WAXS: gradual rise of 0.373 mTorr/hour



MAXS: gradual rise of 0.350 mTorr/hour



A SHARED UPWARD TREND, NOT CONTINUED IRRADIATION

WAXS: 0.049 → 0.052 Torr (+3.0 mTorr, +6.1%).
MAXS: 0.045 → 0.048 Torr (+3.0 mTorr, +6.7%).
Endpoint values are the first and last 15-minute medians.

Both gauges rise in step-like increments of about 0.001 Torr. The long plateaus and correlated samples do not support precise statistical error bars on a fit. The fast shutter remained closed, so this interval adds no Kapton beam exposure; BPM3 near 49 reports upstream flux, not beam reaching the sample.

The data establish increasing indicated pressure in both regions. They do not separate changing leak/outgassing load, pumping performance, temperature or gauge response. The rapid rise beginning around 06:41 is excluded from this fit.